10 - Radio communications

10.1 - Label

10.1.1 - Basic principles and terms

To ensure clear and satisfactory transmission, there are a couple of considerations to be made regarding radio communications. In reality, the quality of the sound obviously depends on many factors, including the distance between the stations, their altitude, the quality of the equipment such as microphones and headphones and above all on the interlocutor. For this reason, it is good to adopt some basic practices before and during a radio call to avoid misunderstandings:

- Make sure that no one is transmitting before transmitting their message (in reality it can only transmit one station at a time, in the case of double transmission the messages are added together creating an annoying whistle on the frequency)
- Be sure not to interrupt an ongoing operational conversation unless absolutely necessary
- Before speaking, recreate the message to be conveyed in your mind so that you do not have doubts or hesitations that would occupy the frequency for nothing
- Avoid hesitation sounds like uh... aaaahm oooh
- Try to speak clearly and punctuating the words
- Press the transmit button fully before speaking and at the end of the transmission, to avoid transmitting only part of the message
- Especially in the event of a frequency change (and in any case in any case) make sure you don't meddle in an ongoing conversation, perhaps waiting a few seconds before transmitting after the change

The term **ROGER** (RECEIVED) should be used to confirm that you have received and understood the message when one of these conditions is not necessary:

- an application for which READ BACK is required,
- a direct affirmative (AFFIRM) or negative (NEGATIVE) response.

The term **STANDBY** simply means wait, I will call back once ready. It does not need readback and does not indicate either an affirmative or a negative answer.

The term **UNABLE** is usually followed by a reason and indicates a condition for which you are unable to execute a request. *IF FOR ANY REASON YOU DON'T THINK YOU CAN COMPLETE THE CONTROLLER'S REQUEST, ANSWER WITHOUT PROBLEMS USING THIS WORD, much better to try again or request something better than to mess up in order to satisfy an inappropriate or wrong request!*

The term **WILCO** (WILL COMPLY) should be used to confirm that you have received the message and to indicate that you will comply with the request or instructions received.

10.1.2 - Order

In standard radio, there is also an order in which a transmitted message must be completed. This order depends on the types of communication you intend to convey and we will not see them all. Generally, however, this rule is followed for the first contact: ***RECIPIENT, SENDER, MESSAGE***

In subsequent communications, however, it is not necessary to include the recipient, SENDER and MESSAGE are sufficient. In a reply, moreover, your callsign can be placed at the end of the reply, according to the order **MESSAGE, SENDER.**

10.1.3 - Readback

A readback is simply a confirmation of an instruction or essential information by repetition to the calling station. This will have to be done by repeating the essential parts in full, and the receiving station will listen to make sure that the message requesting it has been understood correctly. The readback should only be done with some phrases and keywords such as:

- Authorizations to enter, land, take off, cross or perform a counter-runway on any active runway;
- Instructions for parking, taxiing, starting;
- Route authorizations such as directed to navigation points, approaches or standard routes;
- Runways in use, altimeter, instructions for elevation, heading or speed changes, and transition levels;
- Radio frequencies.
- Any other type of authorization must be confirmed in a clear manner that indicates that the instruction has been understood and will be carried out (WILCO).

10.1.4 - Radio checks

A radio check can be used to check the quality and correct functioning of your radio equipment. This is usually done by calling in frequency with RADIO CHECK, and you wait for the response of the other station with a code from 1 to 5, which indicates the quality of your transmission. This scale of 1 to 5 has the following meaning:

1. Unreadable
2. Readable now and then (understandable at times)
3. Readable but with difficulty
4. Readable
5. Perfectly readable

10.1.5 - Flight hierarchy

Let's take a formation of 4 aircraft as a reference. In this line-up we will find the leader (number 1) and the wingmen (2, 3 and 4). The wingmen in turn have a role within the formation:

1. FLIGHT LEADER: navigation, radar, check 6 and external control
2. FLIGHT WINGMAN: navigation and radar support, check 6 and external control
3. ELEMENT LEAD: Radar support, check 6, external control and backup navigation
4. ELEMENT WINGMAN: check 6 and external control

As for the radios, the leader will always be the communications manager for the whole squadron with a few exceptions:

- A wingman is asked
- A wingman is assigned to communicate via radio instead of the leader
- The leader is unable to communicate (due to breakdowns or emergency situations)
- The flight has separated into two distinct elements, so the lead element becomes responsible for its own element.
- Emergency situations

The radio identifier will be the name of the flight (such as Bear, Chevy, etc.) followed by the flight number, followed by the single element. Example: BEAR 1-1 indicates Bear flight number 1, and the personal identifier is that of the leader (1). BEAR 2-1 indicates Bear flight number 2, again indicating the leader. BEAR 2-2 always indicates flight number 2, but the transmitting/receiving element will be number 2.

For intra-flight communications, the leader will identify himself with the callsign of the flight and his own number (ergo 1), while the various wingmen will respond to the various requests of the leader by communicating their number (2, 3 or 4) usually followed by a brevity or a message of understanding (roger, wilco, copy, unable, etc.). Example: BEAR 1 fence in, 2, fenced in, 9.7, 3, fenced in 9.5, 4, fenced in 9.9

10.1.6 - Required Reports

In flight there will be some types of communication that will have to be mandatorily transmitted to the command or to the other aircraft. These messages are:

- Assigned flight level change
- Check-in
- Detection of threats both on the ground and in the air
- Start of attack
- Frequency change

10.1.7 - Examples

- Kobuleti, BEAR 1-1, 4 ships F18, apron south, request taxi to active runway.
- BEAR 1-1, Kobuleti, altimeter 2992, taxi to holding point A for runway 25 via S.
- Altimeter 2992, taxi to holding point A runway 25 via S, BEAR 1-1.

Or

- BEAR 1-1, Altimeter 2992 taxi to holding point A runway 25 via S.
- Kobuleti, BEAR 1-1, radio check.
- BEAR 1-1, Kobuleti, read you 5 by 5.
- BEAR 1-1, Overlord, WP2 overhead report.
- BEAR 1-1, Wilco

Or

- Wilco, BEAR 1-1
- BEAR 1-1, contact 249.000 -> It reads TWO FOUR NINER DECIMAL ZERO
- 249,000, BEAR 1-1

10.2 - Ground flight phases

For this little guide we will rely on 6 maps (available in the DCS kneeboard) of Al Minhad, the GND part (on the ground), the standard VFR procedures of JATF and 3 IFR maps (SID, STAR and ILS approach). These should be used as a reference to better understand this guide but are not, of course, the only cases or communications used. To make it easier to understand, messages transmitted by an aircraft will be highlighted in blue, those transmitted by a controller in orange, and those transmitted by third parties (such as wingmen) in green.

10.2.1 - Starting

During or before the start-up the leader will call the tower to check the correct operation of the radio, to obtain the latest weather information (such as wind, altimeter and any clouds) and to request authorization to start. If an ATIS is present, it is the leader's job to be aware of the weather conditions in the field without requesting them from the controller beforehand. The leader shall also request the departure procedure or VRP desired for their flight plan. Wingmen can control operation via the intra-flight frequency as it would be advisable not to clog the main frequency unnecessarily.

Once the flight has been cleared for start-up, the flight will receive the VFR reporting point or SID assigned for its departure. In case the flight has planned a different exit point, it can request the controller to change the assigned procedure, but this is not guaranteed.

Usually, the airport ATIS frequency will transmit the following information (not necessarily in this order):

- Station/airport name
- Identifier of current information, indicated by a letter (such as R, Romeo)
- Time of meteorological observation (in UTC)
- Runway in use
- Instrument approach in use
- Wind
- Elevation and cloud density, with 0 elevation based on the field
- Visibility
- Temperature and dew point
- QNH/altimeter
- Any other information, such as thunderstorms, the presence of birds or other.

It is usually sufficient to report the letter of the identification to the controller, as it indicates a precise meteorological observation and it is considered obvious that the pilot knows the rest of the transmission.

Example:

This is Kobuleti information A, at time 1610, runway in use 06, ILS approach runway 06, wind 070 at 8, clouds scattered 2000, temperature 15, dew point 9, QNH 1014. This was Kobuleti information A.

The leader can then contact the frequency in use:

GROUND STATION, FLIGHT, radio check

FLIGHT, STATION, read you [BROADCAST QUALITY]

Transmission quality means the code explained in the Radio Checks paragraph.

STAZIONE, FLIGHT, request startup and weather

Or, if an ATIS is available

STATION, FLIGHT, request startup and departure via [VRP or SID], information R (Romeo).

FLIGHT, STATION, startup approved, runway in use [N° RUNWAY], [WEATHER INFO], cleared via [VRP or SID] departure, squawk [SQUAWK] sequential

Startup approved, runway in use [N° RUNWAY], QNH/ALTIMETER, cleared via [VRP or SID] departure, squawk [SQUAWK] sequential, FLIGHT

Or, if an ATIS is available

FLIGHT, STATION, cleared for startup, cleared via [VRP or SID] departure, squawk [SQUAWK] sequential

Startup approved, cleared via [VRP or SID] departure, squawk [SQUAWK] sequential, FLIGHT

Example:

- Al Minhad, BEAR 2-1, radio check
- BEAR 2-1, Al Minhad, read you 5 by 5
- Al Minhad, BEAR 2-1, request startup and departure via Sierra.
- BEAR 2-1, Al Minhad, startup approved, runway in use 08, wind 030 for 5kts, CAVOK, altimeter 2992, cleared via Sierra, squawk 4001 sequential
- Startup approved, runway in use 08, altimeter 2992 (or QNH 1013), departure via Sierra, squawk 4001 sequential, BEAR 2-1

10.2.2 - Taxiing

When the flight is ready to taxi, the leader will call the tower requesting permission to move. In the event that a VFR departure is requested, the communications will be as follows:

GROUND STATION, FLIGHT, NUMBER AND TYPE, ROLE, LOCATION, request taxi

At this point the controller on the ground should respond like this:

FLIGHT, STATION, QNH/ALTIMETER, taxi to holding point [HOLDING POINT] runway [RUNWAY NUMBER] via [TAXIWAY]

Example:

- Al Minhad, BEAR 2-1, 4 F18 strike, apron south, request taxi, departure via November
- BEAR 2-1, Al Minhad, altimeter 2992 (or QHN 1013), taxi to holding point A runway 08 via S
- Altimeter 2992 (or QNH 1013), taxi to holding point A runway 08 via S, BEAR 2-1

At this point the leader will communicate the taxi out to his flight in this way, and the wingmen will respond with an acknowledgement:

CALLSIGN, taxi out.

ELEMENT, taxi out/roger.

Example:

- BEAR 2-1, taxi out
- 2, Roger
- 3, Roger
- 4, Roger

10.2.3 - Take-off

Once you arrive at the designated waiting point and ready for takeoff, the leader will contact the tower and bring back ready:

FLIGHT, HOLDING POINT, ready for departure

It is essential to underline that words such as takeoff, takeoff, landing and landing are only used to authorize aircraft to take off/land and not for departures/arrivals. This is done to avoid any confusion that could lead to runway incursions during landings/takeoffs of other aircraft. In addition to this, any take-off/landing authorization or communications containing the runway in use must be retransmitted with a readback. In the event that this is not done, it is the controller's responsibility to retransmit the communication until a positive readback is received.

At that point the tower will authorize the flight at alignment, take-off or in case of traffic on the runway/approach it will tell to maintain the position:

FLIGHT, cleared for takeoff runway, wind

FLIGHT, line up and wait runway [RUNWAY NUMBER]

FLIGHT, backtrack approved runway [RUNWAY NUMBER]

FLIGHT, hold short [RUNWAY NUMBER]

To which the flight will carry out the readback of the authorization.

At the end of the alignment, each element of the flight will report on the intraflight the status of alignment on the runway to the leader with CALLSIGN, lined up. At this point, the standard take-off procedures are carried out in formation. When an element takes off, it must now communicate on the intraflight (not on the frequency of the tower) CALLSIGN, airborne.

Example:

- BEAR 2-1, ready for departure
- BEAR 2-1, line up and wait runway 08 (ATC)
- Line up and wait runway 08, BEAR 2-1
- BEAR 2-1, cleared for takeoff runway 08, wind 030, 5 knots
- Cleared for takeoff runway 08, BEAR 2-1

(Note that wind readback is not necessary as it is simply information)

10.2.4 – VFR Departures

If VFR procedures are used, the flight does not need communication until it arrives at the authorized VRP (30 seconds before is sufficient):

FLIGHT, [VRP] outbound, 3000 ft

The controller will then authorize the switch to terminal control. If, after 10NM, the tower controller has not yet passed the flight, it is a good idea to contact him and communicate your position.

FLIGHT, contact [STATION 2], [FREQUENCY/CHANNEL]

[FREQUENCY/CHANNEL], FLIGHT

Examples:

- Al Minhad, BEAR 2-1, November outbound, 3000 ft
- BEAR 2-1, contact Overlord, 265.0
- Push 265.0, BEAR 2-1

10.2.5 – IFR Departures

If, on the other hand, instrument procedures are used on departure, the flight must follow the controller's instructions, requesting corrections during the procedure if doubts arise or the limit of one's clearance is being reached. Once the flight has passed from the tower to the approach control for departure, it must communicate:

- Current odds
- Authorized Fee
- Name of the procedure you are following

STAZIONE, FLIGHT, passing [QUOTA] feet, climbing to [QUOTA], on the [NOME SID] departure

10.2.6 - En route

Once you have exited the airport area, the flight will continue its flight plan following the directions of the approach controller (if in use). Please remember to communicate the preferred cruising altitude of the flight being planned (for fuel or other reasons) and your heading (or the point to which you are flying) in order to allow the controller to approve the request and manage traffic in the area:

STAZIONE, FLIGHT, passing [QUOTA], requesting [CRUISE FL] for cruise, direct to/on heading [PUNTO/PRUA]

In any case, before entering the area of operations or when switched to AWACS, the leader will contact him as follows:

STATION, FLIGHT, NUMBER AND TYPE, ROLE, POSITION FROM BULLS, QUOTA, checking-in as fragged

Example:

- Overlord, BEAR 2-1, 4x F18 strike, BULLS 310 for 30, angels 15 for 35, checking in as fragged

Generally, once checked-in, the AWACS will begin the management of the various flights in the operational area by distributing the various roles and targets to the squadrons.

If necessary, the TAC C2 entity may require identification by RAMROD code as follows:

FLIGHT, STATION, authenticate A B

To which the flight must respond:

FLIGHT, I authenticate C

A, B and C in this case are generic letters of a RAMROD that must be pronounced using the standard phonetic alphabet.

FLIGHT, STATION, good authentication

10.2.7 - Check-out

Once the mission is completed or the armaments are exhausted, the leader will report the status to the respective AWACS. This can clearly be of various types depending on the reason for the return based:

- Bingo fuel
- Mayday or Pan Pan
- Winchester
- Mission completed

STATION, FLIGHT, [STATUS] RTB

Examples:

- Overlord, BEAR 2-1, bingo fuel, RTB
- Overlord, BEAR 2-1, Winchester, RTB
- Overlord, BEAR 2-1, mission complete, RTB
- BEAR 2-1, RTB approved, contact Al Minhad on 128.5
- RTB approved, Al Minhad on 128.5, BEAR 2-1

10.2.8 - RTB

Once checked out with the AWACS, the flight will proceed following the route agreed with the radar controller (if present) remembering to communicate any emergency or priority situations, so that we can plan the airspace and manage traffic in the area. In the event that radar or approach controllers are present, it is the leader's responsibility to request descent from cruising level, heading or heading to specific points and communicate a preferable type of approach (always depending on weather conditions).

At the appropriate time, the flight may require drop-off at the airport in this way:

STATION, FLIGHT, ready for descent/request descent

FLIGHT, descend to [FL/ALT]

Descend to [FL/ALT], FLIGHT

At this point, the controller will direct the flight towards the type of approach in use (or chosen by the flight) passing control to other sectors, such as those of the approach. Once in contact with the approach, flights must be ready to carry out the instructions given as precisely as possible. It is important to remember that pilots can express their intentions for landing by communicating what type of approach they prefer between VFR or the various instruments available at the airport.

In the event of an emergency situation with significant damage, you must declare the status of MAYDAY/PAN PAN together with your callsign. It would also be a good idea to include any low-fuel situations in your call, so that you can prioritize flying over landing and avoid moving on to emergency situations.

10.2.9 - VFR approaches

In the event that VFR re-entry procedures are used, the flight will eventually be authorized to descend to the altitude of the nearest VRP (remember, 2000ft AAL) and once in the vicinity of the point it will have to request authorization to return to the airport area (for procedures consult OM-B.2.8). Please note that it is absolutely forbidden to enter the airport area without having obtained the correct authorization.

STATION, FLIGHT, [VRP] inbound for landing

FLIGHT, STATION, cleared to join left/right circuit runway [RWY], report downwind

Or:

FLIGHT, STAZIONE, cleared to the initial for runway [RWY], report at the initial

Or, if the airspace is temporarily full, the flight must be kept outside the zone:

FLIGHT, STATION, hold at [VRP]

Once approaching the initial (if authorized to a jet circuit, the most common) the flight must report its position:

FLIGHT, initial for runway [RWY]

FLIGHT, cleared for left/right/jet circuit runway [RWY], report on downwind/base/final runway [RWY]

To which it is necessary to respond with a readback.

10.2.10 – IFR approaches

No later than about 50 miles from the airport, the flight must request passage to the terminal controller for approach, if it is not passed by the controller currently in contact. The first contact with the controller responsible for the approach must be made in this way:

STATION, FLIGHT, NUMBER AND TYPE, LOCATION (relative to airport or TACAN), [FL/ALT]

At this point, the controller will respond by informing the flight of the runway in use and provide instructions for approaching the airport.

FLIGHT, STATION, ALTIMETER/QNH, proceed/cleared to [POINT/APPROACH], expect [ILS/TACAN/VOR] approach for runway [RWY], descend [FL/ALT]

Subsequently, the controller may respond differently depending on the traffic conditions it is managing and the type of flight it has to control (IFR/VFR). This includes, but is not limited to:

- Vectors: ATC will initially instruct carriers and altitude/FL restrictions to the formation to bring it to a certain point from which to initiate an ILS, TACAN, or VOR approach:

  FLIGHT, STATION, turn left/right heading [HDG], descend [FL/ALT]

  and later:

  FLIGHT, STATION, cleared [ILS/TACAN/VOR] approach runway [RWY]

- Procedural: ATC will instruct the flight to follow a certain arrival procedure (such as a STAR) that will take them to the starting point for the instrument approach in use:

  FLIGHT, STAZIONE, cleared for the [STAR] arrival, descend to [FL/ALT]

It is important to underline that any quota or speed restrictions present along the authorized procedure must be respected, except when expressly communicated by the controller:

FLIGHT, STATION, descend unrestricted to [FL/ALT]

At this point, depending on the approach and once authorized, the flight will report to the controller its position once stabilized on the radial of the initial initial/fix to begin the actual approach. The position should generally be around 10 miles (variable depending on the procedure) and the approach controller will pass the flight to tower control for landing

FLIGHT, established on [APPROACH] runway [RWY], [DISTANCE]

FLIGHT, contact tower on [FREQ]

10.2.9- Landing

Once the circuit or instrument approach has begun, the tower will authorize you to land (if available), make you continue without authorizing you to land, or ask you to return to a certain position.

FLIGHT, STATION, report on final/downwind, runway [RUNWAY NUMBER]

FLIGHT, STATION, continuous approach

FLIGHT, STATION, wind, cleared to land

Once the position is reached, the flight will tell the tower that it has reached it.

FLIGHT, on final/downwind, runway [RUNWAY NUMBER]

At this point, being now close to the runway, the tower will authorize the flight upon landing or order a go-around.

FLIGHT, STATION, wind, cleared to land

FLIGHT, go around!

In the event of a go-around, the airport procedures or the controller's instructions will be followed and then a new approach will be started. In both cases, it is essential and mandatory to carry out a readback as soon as possible.

10.2.10 - Taxi in

At this point the flight has landed and is waiting to receive instructions to be able to clear the runway (a maneuver to always be carried out as quickly as possible as there may be other flights landing) and for the return to the parking lot. The tower will then call like this:

FLIGHT, vacate L/R via [EXIT], taxi to [PARKING] via [TAXIWAY]

Except for special cases or requests (such as stopping taxiing at certain intersections) it will therefore be the last radio communication of the flight with the controller on the ground. You do not need to request shutdown before shutting down the aircraft.

10.3 - Summary for each phase

10.3.1 - Starting

BEAR 2-1	Al Minhad Ground
Al Minhad GND, BEAR 2-1 for radio check	
	BEAR 2-1, Al Minhad GND, read you 5 by 5
Al Minhad GND, BEAR 2-1, request startup and departure via ECHO	
	BEAR 2-1, Al Minhad GND, startup approved, runway in use 09, altimeter 2992. Cleared departure via ECHO, squawk 5010 sequential.
Startup approved, runway in use 09, altimeter 2992 (or QNH 1013), cleared departure via ECHO, squawk 5010 sequential, BEAR 2-1	

10.3.2 - Taxiing

Al Minhad GND, BEAR 2-1, 4 F18 strike, apron 2, request taxi.	
	BEAR 2-1, taxi to holding point A runway 09 via B1, H
Taxi to holding point A runway 09 via B1, S	
	BEAR 2-1 contact Al Minhad Tower on 248.2
248.2, BEAR 2-1	

10.3.3 – Take-off and departure

BEAR 2-1	Al Minhad Tower
Al Minhad TWR, BEAR 2-1, ready for departure	
	BEAR 2-1, AL Minhad TWR, line up and wait runway 09
Line up and wait runway 09, BEAR 2-1	
	BEAR 2-1, wind 030, 5 knots, cleared for takeoff runway 09.
Cleared for takeoff runway 09 7, BEAR 2-1	
Al Minhad TWR, BEAR 2-1, ECHO outbound	
	BEAR 2-1, contact Al Minhad APP on 230.4
230.4, BEAR 2-1	

10.3.4 - En route

BEAR 2-1	Al Minhad Approach
Al Minhad Approach, BEAR 2-1, ECHO outbound 3000ft, requesting FL320 for cruise	
	BEAR 2-1, Al Minhad, radar contact, climb FL200 initially
Climb FL200, BEAR 2-1	
	BEAR 2-1 contact Emirates Control on 124.375
124.375, BEAR 2-1	
BEAR 2-1	**Emirates Control**
Emirates Control, BEAR 2-1, climbing FL200	
	BEAR 2-1, Emirates Control, radar contact, climb and maintain FL320
Climb and maintain FL320, BEAR 2-1	

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	BEAR 2-1 radar service terminated, contact Overlord 1-1 on Orange 31
Orange 31, BEAR 2-1	
BEAR 2-1	**Overlord 1-1**
Overlord 1-1, BEAR 2-1, checking in as fragged, alpha check bullseye 200 for 30	
	BEAR 2-1, Overlord 1-1, radar contact, alpha check sweet

10.3.5 - Check-out

Overlord, BEAR 2-1, Winchester, RTB	
	BEAR 2-1, RTB approved, contact Emirates CTL on 124.375
RTB approved, 124.375, BEAR 2-1	

10.3.6 - RTB and approach

10.3.6.1 – Visual VFR

BEAR 2-1	**Emirates CTL**
Emirates CTL, BEAR 2-1, 4 Hornet, 80NM inbound ECHO for landing, FL320	
	BEAR 2-1, Emirates CTL, identified, descend FL100
Descend FL100, BEAR 2-1	
	BEAR 2-1, contact Minhad APP on 230.4
230.4, BEAR 2-1	
BEAR 2-1	**Minhad APP**
Minhad APP, BEAR 2-1, descending FL100, ECHO inbound for landing	
	BEAR 2-1, Minhad APP, proceed to ECHO, descend 2000ft on altimeter 29.80 (QNH 1009)
Altimeter 29.80 (QNH 1009), proceed to ECHO, descent 2000ft, BEAR 2-1	
BEAR 2-1, ECHO inbound, 2000 feet	
	BEAR 2-1, contact Minhad TWR on 248.2 BEAR 2-1, contact Minhad TWR on 248.2
248.2, BEAR 2-1	
BEAR 2-1	**Minhad TWR**
Minhad TWR, BEAR 2-1, ECHO inbound, 2000ft	
	BEAR 2-1, Minhad TWR, report initial runway 09
Wilco, BEAR 2-1	
BEAR 2-1, initial runway 09	
	BEAR 2-1, cleared for left circuit runway 09, report on final
Cleared for left circuit runway 09, wilco, BEAR 2-1	

10.3.6.2 – Instrumental IFR

BEAR 2-1	**Dubai CTR**
Dubai CTL, BEAR 2-1, 4 Hornet, 80NM inbound MIN TACAN, FL320	

	BEAR 2-1, Dubai CTL, turn right heading 220, descend FL100, expect ILS approach runway 09
Descend FL100, right heading 220, BEAR 2-1	
	BEAR 2-1, turn right heading 150, descend 2200ft on altimeter 29.80 (QNH 1009), cleared for ILS approach runway 09
Turn right heading 150, descend 2200ft on altimeter 29.80 (QNH 1009), cleared for ILS approach runway 09, BEAR 2-1	
	BEAR 2-1, 10nm on final, contact Minhad TWR on 248.2
248.2, BEAR 2-1	
BEAR 2-1	**Minhad TWR**
Minhad TWR, BEAR 2-1, established ILS runway 09, 8nm on final	
	BEAR 2-1, Minhad TWR, continue approach,report 4nm on final
Wilco, BEAR 2-1	

10.3.7 - Landing

BEAR 2-1	Minhad TWR
BEAR 2-1, on final runway 09	
	BEAR 2-1, wind 030 5kts, cleared to land runway 09
Cleared to land runway 09, BEAR 2-1	
(if a go-around from the tower is called)	BEAR 2-1, go around!
Going around, BEAR 2-1	

10.3.8 - Taxi in

	BEAR 2-1, vacate left on F and contact Minhad GND on 290.6
Vacate left on F, 290.6, BEAR 2-1	
BEAR 2-1	**Minhad GND**
Minhad GND, BEAR 2-1, vacating via F	
	BEAR 2-1, Minhad GND, taxi to apron L via H, L2
Taxi to apron L via H, L2, BEAR 2-1	

10.4 - Flight phases for carrier operations

In carrier operations there are some differences compared to land operations. In the real world, CASE I take-off and landing operations take place in what is called zip-lip, i.e. no calls once passed with the tower frequency. This is to leave the frequency free to the LSO or emergency calls, as the whole procedure is in sight, each aircraft must autonomously range from the others and manage its approach based on what is happening around it.

In addition, tactical callsigns are not used in communications during carrier procedures, but the aircraft number (such as 2-1, 947, etc.).

10.4.1 - Starting

As far as starting on aircraft carriers is concerned, there is no need to ask for your authorization, which is only communicated to everyone on deck through the 5MC, or the bridge's loudspeaker communication system.

10.4.2 - Taxiing

On aircraft carriers there is no need to request taxiing authorization, you just have to announce your number by radio with up and ready. This is to signal one's intention to move on the bridge.

SIDE NO, up and ready

10.4.3 - Take-off

There is no need to request take-off authorization from the catapult to which you are assigned and once you have taken off there are 3 different cases based on the type of CASE conditions:

- CASE I: Zip-lip conditions, so no call needed
- CASE II: same as case I, with the only difference being that you need to report if you are still in IFR conditions once you reach 18000 feet by reporting popeye

SIDE NO, popeye

- CASE III: return airborne after launch, report passing 2.5 passing over 2500 feet, report arcing at intercept of the 10NM arc, report stable outbound established outbound on the outbound radial, bring back kilos when ready to continue the mission. Popeye conditions should also be reported if still in cloud at 18000 feet and on top with altitude once free of clouds.

SIDE NO, airborne

SIDE NO, passing 2.5

SIDE NO, arcing

SIDE NO, established outbound

SIDE NO, kilo

SIDE NO, on top, angels XX

SIDE NO, popeye

10.4.4 - En route

No communication is required to switch frequencies to AWACS under CASE I or II conditions. In case of CASE III conditions, instructions will be received once you have communicated that you are ready to continue the mission. If there is no human controller you can go directly to the next frequency:

SIDE NO, kilo

SIDE NO, STATION, copy, push [FREQUENCY/CHANNEL]

10.4.5 - Check-out

Please refer to chapter *3.2.5 – Check-out*.

10.4.6 - RTB and Approach

Once checked out with the route controllers and before 50NM from the ship, the leader (or individuals/couples in case of CASE II/III conditions) will call the Marshal to check in and receive instructions for approaching the aircraft carrier.

MARSHAL, SIDE NO, holding hands with [GREGARI], marking Mom's [RADIALE] for [DISTANZA], angels [QUOTA], low state [FUEL]

In the event of a call for a single approach, the holding hands section can be omitted and it is sufficient to report the status FUEL.

At this point, in case of **CASE I conditions** , the Marshal will reply with:

SIDE NO, mother's weather is [VISIBILITA'], [NUVOLE], altimeter [QNH]. CASE I recovery expected BRC is [HEADING SHIP]. Report see me at 10

To which you must answer only with your aircraft number. Once the ship is in sight and at 10NM, the pilot must report:

SIDE NO, see you at 10

Followed by the Marshal's response:

SIDE NO, update state, contact tower

And from the pilot's answer:

SIDE NO, state/low state [FUEL]

From this moment on, the zip-lip phase would begin, i.e. without communications, in which the flight/pilot enters the stack and manages himself until he is in sight of the LSO at 3/4 of a mile. In DCS, however, the automatic ATC will communicate in this way:

Tower, SIDE NO, overhead angels [QUOTA], holding hands with [GREGARI], state/low state [FUEL]

The rook will answer us once we are free to start the circuit and leave the stack like this:

SIDE NO, tower, Roger. BRS is [HEADING SHIP], signal is charlie

To which we will only respond by answering with our aircraft number.

In the event of a **case ii/iii approach,** the Marshal's response will be as follows:

SIDE NO, CALLSIGN NAVE marshal, CASE II/III recovery, CV-1 approach, expected BRC is [HEADING SHIP], altimeter [QNH]. SIDE NO, marshal mother's [RADIAL] radial, [DISTANCE] DME, angels [ODD]. Expected approach time is [MINUTI].

To which the readback must be carried out in the following way:

SIDE NO, marshal on the [RADIALE], for [DISTANCE] DME, angels [QUOTA]. Expected approach time [MINUTES]

The marhal will then answer us:

SIDE NO, readback correct

At this point the pilot will move to the assigned altitude and towards the stack according to the approach procedures, reporting once the assigned altitude has been reached:

SIDE NO, established angels [QUOTA]. State [FUEL]

To which the marshal will respond with:

SIDE NO, roger, state [FUEL]

Once the time has come to start the approach and leaving the marshal fix, the pilot will report:

SIDE NO, commencing, [QNH], state [FUEL]

With the consequent answer of the marshal:

SIDE NO, radar contact [DISTANZA] miles, expected final bearing [BEARING]

To which the pilot will respond with his aircraft number. Once stable on the descent, the Marshal will give instructions to contact the approach:

SIDE NO, switch approach

To which the pilot will respond with his aircraft number. At this point, check-in must be carried out with the approach:

SIDE NO, checking in, [DISTANCE] miles, state [FUEL]

With answer:

SIDE NO, final bearing [BEARING]

To which the pilot will respond with his own number. Once you reach 5000 feet, you must communicate that you have reached the platform:

SIDE NO, platform

With Approach Response:

SIDE NO, roger

Once the final bearing has been intercepted, the controller will instruct us to follow the ICLS indications:

SIDE NO, fly bullseye

To which the pilot will respond with his own number. At this point during a. CASE II the pilot will report that he has the ship in sight at 10 miles and will follow CASE I procedures. At about 6-8 miles the controller will report:

SIDE NO, final radar contact, [DISTANCE] miles

Once the 6 miles have been reached, the controller will ask you to check the ACLS system indications:

SIDE NO, ACLS lock on, [DISTANCE] miles, say needles

To which the pilot must respond by indicating the location of the ACLS circle on the HUD:

SIDE NO, up and on (or the relative position)

The controller will respond with:

SIDE NO, concur, fly mode 2

That is, the ACLS mode (2 stands for manual). When the pilot is approaching the glidepath (approximately 4 miles) the controller will call:

SIDE NO, approaching glidepath

To which the pilot replies with his own number.

10.4.7 – Landing

During a CASE I in zip-lip conditions the ball call is not necessary, but in DCS this is not always true. Once I reach the groove (or 3/4 of a mile) the call must be made:

SIDE NO, [AIRCRAFT] ball, state [FUEL]

To which the LSO will respond with Roger ball and begin to provide directions for the landing.

During a case III the controller will call the pilot once they reach 3/4 of a mile asking him to ask for the ball:

SIDE NO, [GLIDE] [LOC], 3/4 of a mile, call the ball

To which the pilot replies:

SIDE NO, [AIRCRAFT] ball, state [FUEL]

To which the LSO will respond with roger balls and begin to provide directions for the landing.

Once landed it is not necessary to make any calls for taxiing or parking, the aircraft will follow the predetermined route to park without any voice communication (in reality the yellow jackets visually manage the movement of the aircraft on the deck).